MCD411A B. Tech. Project

Progress Report 2 (22-Aug-2026 - 28-Aug-2026)

Submission Due: 28-Aug-2026, 06:00 PM

Title of the project: Neural Network Based KKT Solver

Tick (✔) Area of the project:

Design	Industrial ✔	Production	Thermofluids

Student Name with Entry number:

Student Name	Sahitya Rankawat	Manav Gupta	Jeet Anand
Entry number	2023ME21131	2023ME20733	2023ME20874

Supervisor(s) Name:	Kartikey Sharma

Progress/ Activities in the current week: *(3-5 bullet points)*

- We went through the research papers to understand how KKT conditions can be used to train a neural network to solve optimization problems without needing labeled training data.
- We set up a working pipeline in Pyomo to easily define optimization problems with their variables, goals, and constraints.
- We connected the HiGHS solver to test the pipeline and to solve our Pyomo problems and successfully extracted both the optimal answers (x*) and their constraint multipliers (λ*).
- We wrote a Python script to automatically extract the algebraic KKT equations directly from the Pyomo model.
- We tested our extracted KKT equations using the solver's outputs and confirmed that the mathematical error is practically zero, proving our pipeline is ready for the neural network phase.

Declaration:

I/We have met our supervisor on 24th Aug **(Date) at** Online (mail) **(Time) and discussed the report before submission.**

Sahitya	Manav	Jeet Anand
(Student 1 Signature)	**(Student 2 Signature)**	**(Student 3 Signature)**

Performance Assessment: {To be filled in by the Supervisor}

Unsatisfactory		Incremental		Satisfactory	

(Supervisor Signature)

We have agreed on bi-weekly meets with the Professor, hence this week's report does not contains the signatures

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